

SAMARKAND, Uzbekistan

WMO: 386960

Lat: 39.701N

Lon: 66.984E

Elev: 2224

StdP: 13.55

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.7	19.2	6.7	8.4	19.3	11.8	10.9	28.3	21.4	38.6	18.7	38.8	4.7	140	0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.4	98.5	66.1	95.8	65.4	93.5	64.7	69.3	91.6	67.8	90.2	66.5	88.8	7.7	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.0	90.1	77.5	59.9	83.6	76.0	58.4	79.1	75.1	34.8	91.2	33.5	90.5	32.4	89.1	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.0	18.4	15.7	9.8	103.4	7.5	1.7	4.4	104.6	0.0	105.6	-4.2	106.6	-9.7	107.8		
(5)			9.0	72.6	7.5	2.2	3.6	74.2	-0.8	75.5	-5.0	76.7	-10.4	78.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.8	36.8	39.3	49.7	59.2	69.2	77.1	80.1	78.0	69.8	58.8	47.4	39.3		
	DBStd	17.17	9.22	10.06	9.30	8.42	6.21	4.78	4.25	4.52	5.81	7.79	8.99	8.83		
	HDD50	1387	416	317	119	21	0	0	0	0	0	20	148	346		
	HDD65	3881	874	721	480	207	30	1	0	1	23	216	530	798		
	CDD50	4611	7	16	108	297	596	814	932	869	593	293	72	14		
	CDD65	1630	0	0	5	34	161	365	467	405	166	24	3	0		
	CDH74	19927	0	2	60	369	1668	4519	6122	4959	1838	353	37	0		
	CDH80	9913	0	0	10	95	600	2318	3425	2656	717	86	6	0		
(14) Wind	WSAvg	6.2	5.4	6.2	6.7	7.0	6.9	6.8	6.6	6.2	6.5	5.6	5.2	5.4		
(15) Precipitation	PrecAvg	15.4	1.8	2.1	2.9	2.4	1.4	0.3	0.1	0.1	0.1	0.6	1.7	1.9		
	PrecMax	22.8	3.4	4.7	5.3	5.6	4.2	0.8	0.7	0.7	0.5	1.7	4.9	4.5		
	PrecMin	7.1	0.6	0.6	1.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0		
	PrecStd	3.8	0.7	1.1	1.1	1.2	1.2	0.3	0.2	0.2	0.1	0.5	1.2	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.6	69.8	80.7	87.6	93.3	99.8	102.6	100.3	94.3	87.7	79.0	66.9		
		MCWB	46.7	50.6	55.0	61.1	64.6	66.1	67.2	66.8	64.0	60.4	55.3	47.7		
	2%	DB	58.3	62.6	73.8	82.5	89.6	96.4	99.5	97.0	91.0	82.0	71.5	61.1		
		MCWB	45.4	48.0	54.0	60.6	63.2	65.1	66.6	65.9	63.0	57.7	52.4	46.8		
	5%	DB	53.8	57.6	69.4	78.4	86.2	93.6	96.6	94.9	87.7	77.5	66.1	56.8		
		MCWB	43.2	46.1	52.9	59.0	62.5	64.5	65.7	65.2	61.6	56.4	50.8	44.9		
	10%	DB	49.8	53.6	64.5	73.7	83.1	91.3	93.6	92.3	84.2	73.7	61.3	52.3		
		MCWB	41.8	44.3	51.4	57.3	61.4	63.9	65.0	64.2	59.9	55.1	49.2	42.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.2	52.7	58.6	65.6	68.4	70.9	71.5	71.3	67.7	62.9	57.3	50.6		
		MCDB	58.9	64.4	73.4	79.6	85.6	91.7	94.4	92.6	87.1	81.5	69.9	60.4		
	2%	WB	46.7	49.9	56.2	62.8	65.8	68.2	69.2	68.6	64.7	59.5	54.6	48.0		
		MCDB	55.5	58.8	69.0	76.6	83.2	90.2	93.2	90.8	86.0	77.0	67.4	58.5		
	5%	WB	44.5	47.6	54.1	60.7	64.3	66.6	67.8	67.0	62.7	57.8	52.3	45.9		
		MCDB	51.7	55.8	65.8	74.5	81.8	88.4	91.6	89.7	84.0	74.3	63.9	54.9		
	10%	WB	42.3	45.4	52.2	58.8	62.8	65.3	66.5	65.6	60.9	56.1	50.1	43.7		
		MCDB	48.6	52.6	62.9	70.2	79.5	86.7	90.0	88.5	81.5	71.1	59.8	50.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.5	17.0	18.9	20.2	22.8	25.5	26.4	27.1	26.4	24.2	19.3	16.6		
		MCDBR	21.7	23.3	25.5	26.5	26.8	29.2	29.6	30.2	30.2	29.6	26.6	23.3		
		MCWBR	12.4	12.9	12.3	12.7	11.5	11.7	11.0	11.4	13.0	13.6	13.2	13.3		
	5% WB	MCDBR	19.8	20.4	22.7	22.7	24.4	26.5	28.0	28.8	28.5	26.3	23.2	21.1		
		MCWBR	12.0	12.3	12.2	12.2	11.6	11.6	11.5	11.9	13.3	13.1	12.4	12.8		
(40) Clear-Sky Solar Irradiance	taub		0.355	0.398	0.464	0.482	0.475	0.467	0.448	0.427	0.412	0.450	0.412	0.357		
	taud		2.187	2.073	1.938	1.930	1.989	2.012	2.064	2.111	2.099	1.949	2.035	2.160		
	Ebn at Noon		248	252	247	253	258	259	263	266	261	229	221	235		
	Edh at Noon		33	43	54	58	56	55	52	48	45	48	38	31		
(44) All-Sky Solar Radiation	RadAvg		588	827	1187	1651	2110	2430	2419	2190	1763	1185	714	528		
	RadStd		74	110	111	174	117	70	78	68	55	86	90	71		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+0.76	N/A	N/A	N/A	N/A	+176	+140
(47) Regional (8 neighbors)	N/A	N/A	N/A	+0.68	N/A	N/A	N/A	N/A	N/A	+106

Nomenclature: See separate page